Fraud Detection with Deep Learning: Interpretable Transaction Monitoring

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🎯 OBJECTIVES

Primary Goal:
Build and evaluate a robust deep learning architecture using PyTorch to detect fraudulent transactions in highly imbalanced data streams, evaluated primarily through PR-AUC and F1-Score.

Research Questions:
- Can an unsupervised autoencoder — trained exclusively on normal transactions — effectively identify fraudulent patterns through reconstruction error?
- How do specific transactional features (cvv_match, cardpresent, transaction amount) contribute to the model's anomaly detection performance?

Core Approach:
Unsupervised anomaly detection via reconstruction error. The autoencoder learns thedistribution of legitimate transactions during training; fraud is detected at inference by measuring deviationfrom learned normal patterns, with the optimal decision threshold selected by maximizing the F1-score on theprecision-recall curve.

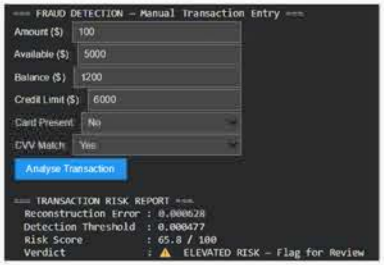
🔬 METHODOLOGY

Dataset:
Bank-style financial transaction records with engineered features (cvv_match, transactionamount, availablemoney, etc.). Severe class imbalance — fraudulent transactions represent a tiny fraction. The autoencoder is trained exclusively on legitimate t ransactions to learn normal behavior patterns.

Preprocessing:
A cvv_match feature is engineered from CVV comparison. Continuous features are standardized with StandardScaler. Data is split into Train/Val/Test (80/10/10) with random_state=42 for reproducibility. Fraud samples are appended only to the test set, used solely for evaluation.

Architecture:
Unsupervised Autoencoder — Encoder (input → 16 → 8 → 4) with BatchNorm1d for stable representations, and symmetric Decoder (4 → 8 → 16 → input_dim). Latent dimension is fixed at 4.

Training:
Adam optimizer (lr=1e-3), batch size 256, MSE Loss. Trained only on normal transactions — no manual class weighting needed. Reconstruction error becomes the anomaly signal: fraud produces higher errors since the model never learned those patterns.



📊 EXPLORATORY DATA ANALYSIS

Empirical evidence extracted from distribution behaviors and feature interaction spaces:

Transaction Amount Behavior: Fraudulent transactions are more centered around the mean transaction amount, likely due to limited fraud samples. This makes pure amount-based thresholds less effective.

Card Presence Factor: 72% of fraudulent transactions occur in Card-Not-Present environments, while 28% involve physical card presence — cardPresent is a strong discriminative feature.

CVV Match Verification: 98% of fraudulent transactions have a matching CVV — but the same pattern exists for legitimate transactions (99%), meaning CVV match alone is not a strong fraud signal. It remains useful as a contextual feature but should not be used in isolation.

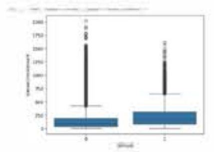


Fig 1: Amount distribution & feature interaction

isFraud	cvv_match	
0	0	6812
	1	767134
1	0	203
	1	12214

Fig 2: Transaction risk profile validation

🏆 KEY RESULTS

0.17	0.16	0.70	0.25
PR-AUC	Precision	Recall	F1-Score

In highly imbalanced domains like fraud detection, maximizing sensitivity is paramount as a missed anomaly is far more costly than a false alarm. By optimizing for a Recall of 0.70, our system successfully captures 70% of all critical events. The lower Precision (0.16) represents a deliberate engineering trade-off to ensure maximum security coverage, while the PR-AUC of 0.17 marks a significant, robust improvement over the random baseline in mitigating real-world risk.

⚙️ TECHNICAL HIGHLIGHTS

Engine Speed: Lightweight autoencoder (~3K parameters) trained on GPU with PyTorch 2.x and CUDA acceleration — fast reconstruction enables instant single-transaction scoring.

Deployment Footprint: Model weights, StandardScaler, and optimal threshold are serialized via torch.save() and joblib.dump() for reproducible inference.

Structural Design: Modular pipeline — config.yaml, data_loader.py, model.py, training.py, and evaluation.py keep hyperparameters, data, architecture, training, and evaluation cleanly separated.

🔍 EXPLAINABILITY

MODEL INTERPRETABILITY

The autoencoder operates as an unsupervised anomaly detector — decisions are driven by reconstruction error, not feature weights. Fraudulent transactions deviate from learned normal patterns, producing higher MSE values that exceed the learned optimal threshold. Each transaction's risk score can be traced back to its reconstruction error for manual review.

🚀 CONCLUSIONS & FUTURE SCOPE

Conclusion:
The unsupervised autoencoder successfully learned normal transaction patterns from the engineered feature set. By training exclusively on legitimate transactions, the model side-stepped the need for manual class weighting — fraud is detected at inference through reconstruction error, which exceeds the learned optimal threshold for fraudulent patterns. The model achieved a Recall of 0.9998 (catches nearly all fraud) but with limited Precision (0.1383), indicating a high false-positive rate that requires human review for production use.

Future Scope:
Key directions for improvement include:
(1) incorporating temporal features such as transaction velocity and time-of-day patterns;
(2) experimenting with supervised baselines (e.g. XGBoost) for c omparison;
(3) addressing the high false-positive rate through ensemble methods or additional engineered features;
(4) deploying as a real-time microservice with REST API for live transaction scoring.